00 The General Species Balance Equation

Why are we learning this?

The General Balance Equation is the foundation of nearly everything we will do in CBE 3610.

Every reactor design equation in this course comes from applying the General Balance Equation to a particular reactor under a particular set of assumptions.

By the end of today's lesson, you should understand:

- How to write a balance on a species.
- What generation and consumption mean.
- Why CSTR and PFR equations look different.
- How assumptions simplify the balance equation.

Course Roadmap

Introduction to CRE
↓
►General Balance Equation◄
↓
Common Reactor Models
(CSTR, Batch, PFR)
↓
Reaction Progress
(Conversion)
↓
Graphical Reactor Design
(Inverse Rate/Levenspiel Plots)
↓
Rate Laws
(Kinetics)
↓
Equilibrium
(Reversible Reactions)

↓

Equilibrium Conversion
(Thermodynamic Limitations)

↓

One Reaction: Solving in terms of Reaction Progress

↓

One Reaction: Reactor Design
(CSTR, Batch, PFR, PBR)

↓

Catalytic Reactions

↓

Numerical Methods
(Python)

↓

Energy Balances

↓

One Reaction: Non-Isothermal Reactor Design

↓

Multiple Reactions

↓

Reactor Safety

↓

Challenging the Well-Mixed Assumption
(Residence Time Distributions)

↓

Challenging the Elementary Rate Law Assumption
(Reaction Mechanisms)

Today's location: General Balance Equation

Big Ideas

The balance equation is simply an accounting statement.

Everything that happens inside a reactor must be accounted for.

What we have later = What we started with + what entered - what left + what was generated

The mathematics can become complicated.

The accounting never changes.

Recall the General Balance Equation:

Steady state system:

Batch system:
The General Balance Equation in CRE:

- Written <u>over</u> a

- Written <u>for</u>

- In CRE, generation and consumption terms are

  - Consider species *j*. If species *j* is net consumed:

  - If species *j* is net generated:

Equation Selection Guide

Students often ask: "How do I know which balance equation to use?"

Start here:

Am I writing a balance?
↓
What reactor do I have: Batch? CSTR? PFR?
↓
Different assumptions lead to different forms of the same balance equation. The equations may look different. The underlying balance is the same.

Common Student Confusion: Generation versus Consumption

Students frequently worry about signs.

Remember:

- Consumed species → generation term is negative
- Produced species → generation term is positive

This is an accounting decision.

The chemistry tells us what is happening.

The sign keeps the bookkeeping correct.

KEY POINT

This is not a "memorize the sign" issue.

This is an accounting issue.

Ask yourself: Is species *j* being created or destroyed?

Then choose the sign that correctly accounts for that behavior.

Let's derive the Mole Balance for species *j*:

Accumulation = In - Out + G_j

How we treat the G_j term depends on what is happening with the system variables.

- What system variables are relevant for CRE?

- If all the system variables are <u>uniform</u>

  - The system is said to be

- G_j =
  - r_j has units of
- Else:

Decision Point

At this stage we must decide:

Are the reactor properties uniform?

Examples of relevant variables:

- Concentration
- Temperature
- Composition
- Reaction rate

If these variables are approximately uniform:

☑ Lumped analysis

If these variables vary significantly with position:

☑ Differential analysis

This decision determines how we evaluate the generation term.

How Do I Know Which Situation I Have?

Are conditions the same everywhere in the reactor?

YES	NO
↓	↓
Uniform system	Spatial variation
(Treat like a well-mixed tank)	(Must integrate the reaction rate over the reactor volume)

Why This Matters Later

By the end of the semester:

- CSTR design equations
- PFR design equations
- Batch reactor balances
- Energy balances

will all come from decisions we make on this page.

This is one of the most important pages in the entire course.

Examples of systems where the variables are non-uniform:

Most Important Takeaway

The General Balance Equation is the foundation of every reactor design equation in this course. All reactor mole balances are obtained by:

1. Starting with the General Balance Equation
2. Writing a balance on a species
3. Making assumptions appropriate for a particular reactor

Examples:

- Batch reactor → no flow in or out
- CSTR → perfectly mixed
- PFR → variables change with position

Coming Next

Today:

- General Balance Equation

Next:

- Batch reactor mole balance
- CSTR mole balance
- PFR mole balance

Notice that the only thing changing is how we evaluate the accumulation, inlet/outlet, and generation terms.